

Collection of plots

Airas Group

RStudio, PBC

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Dot-and-whisker plot

Dot-and-whisker plots visually summarize regression results by showing the point estimate (dot) and confidence interval (whiskers) for each predictor. They are useful for quickly comparing effect sizes and significance across multiple variables in a single, clean plot.

- We start from running logistic regression model
- We obtain estimate (with p-values, confidence intervals) for each of the regressor in the model

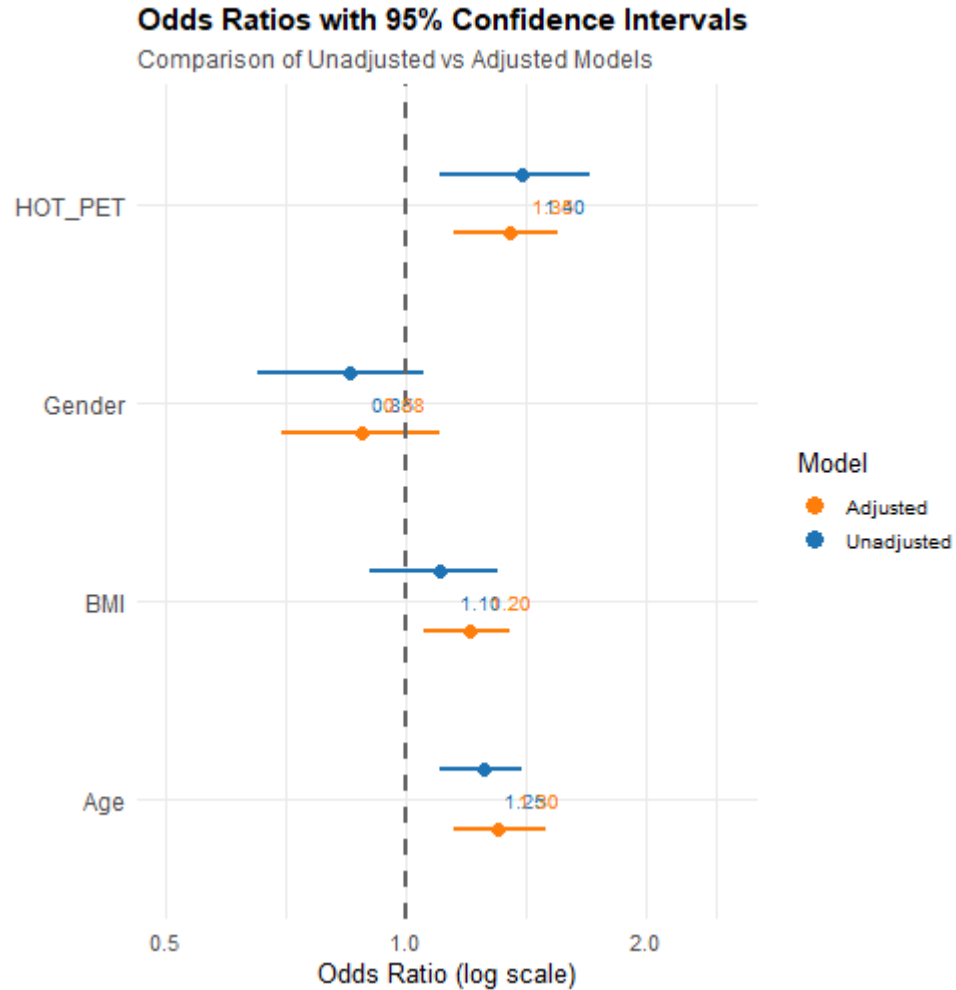
##	Predictor	OR	CI_low	CI_high	p_value	model
## 1	Age	1.25	1.10	1.40	0.010	Unadjusted
## 2	Age	1.30	1.15	1.50	0.020	Adjusted
## 3	Gender	0.85	0.65	1.05	0.040	Unadjusted
## 4	Gender	0.88	0.70	1.10	0.030	Adjusted
## 5	BMI	1.10	0.90	1.30	0.120	Unadjusted
## 6	BMI	1.20	1.05	1.35	0.100	Adjusted
## 7	HOT_PET	1.40	1.10	1.70	0.001	Unadjusted
## 8	HOT_PET	1.35	1.15	1.55	0.002	Adjusted

Now we need to arrange the data before plotting

From the model output (logistic regression) we need to run this:

```
dw_data <- results_clean %>%  
  
  rename(  
    estimate = OR,  
    conf.low = CI_low,  
    conf.high = CI_high,  
    term = Predictor  
  ) %>%  
  mutate(  
    model = factor(model, levels = c("Unadjusted", "Adjusted")),  
    term = fct_rev(factor(term)),  
    label = sprintf("%.2f", estimate),  
    sig = ifelse(p_value < 0.05, "Significant", "Not significant")  
  )
```

And on the `dw_data` arranged for the plot, we run the `dwplot` function:



Mosaic plot

Mosaic plots are used to visualize contingency tables. The size of each tile represents the proportion of observations in each group-category combination, allowing for quick comparison of relative frequencies.

We start from a contingency table as the following one:

```
mat <- matrix(c(73, 27, 23, 77),
              nrow = 2,
              byrow = TRUE,
              dimnames = list(
                Group = c("PF%< 2.12", "PF% >= 2.12"),
                Category = c("Low", "High")
              ))
mat
```

##		Category	
##	Group	Low	High
##	PF%< 2.12	73	27
##	PF% >= 2.12	23	77

Next we run the mosaicplot function:

